

Lecture 1: Complex Number

$i^2 = -1$
① Imaginary unit $i: i^2 = -1$
A complex number has rectangular form: $z = x + iy, x, y \in \mathbb{R}$
where $\text{Re } z = x$ and $\text{Im } z = y$ (real part & imaginary part)
ex. Write $(2+3i)(1-4i)$ in rectangular form:
 $(2+3i)(1-4i) = 2 + 3i - 12i - 12 = -10 - 9i$

② Conjugate and Modulus:
let $z = x + iy$ then its conjugate $\bar{z} = x - iy$.
Remark: ① $\text{Re } z = \frac{z + \bar{z}}{2}$ $\text{Im } z = \frac{z - \bar{z}}{2i}$
② $z\bar{z} = x^2 + y^2 = |z|^2$

Modulus: $|z| = (\bar{z}z)^{1/2} = |x + iy|$
Useful rules: $\begin{cases} \overline{z+w} = \bar{z} + \bar{w} \\ \overline{zw} = \bar{z}\bar{w} \\ \overline{1/z} = 1/\bar{z} \end{cases}$
ex. $z = 3 + 4i$, compute $\bar{z}$, $|z|$, $z\bar{z}$ and $\bar{z}\bar{z}$
① $\bar{z} = 3 - 4i$, ② $|z| = \sqrt{3^2 + 4^2} = 5$
③ $z\bar{z} = (3+4i)(3-4i) = 6$ ④ $\bar{z}\bar{z} = (3-4i)(3+4i) = 25$

③ Division: $\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$
ex. Write $\frac{3+i}{1-4i}$ in rectangular form
 $\frac{3+i}{1-4i} \cdot \frac{1+4i}{1+4i} = \frac{3+11i-4-16i}{1+16} = \frac{-3-15i}{17} = -\frac{3}{17} - \frac{15}{17}i$

④ Geometric Representation:
 $z = x + iy$

 $r = |z|$, then $x = r \cos \theta, y = r \sin \theta$
 θ is an argument of z , written arg z (θ is not unique!)
 $\Rightarrow z = r(\cos \theta + i \sin \theta)$ "polar form"
ex. Convert $z = -1 + i\sqrt{3}$ to polar form using the principal argument.
let $r = |z| = \sqrt{1+3} = 2$
 $z = r(\cos \theta + i \sin \theta)$ where $\theta = \arctan(-\sqrt{3}) = -\frac{\pi}{3}$
 $\Rightarrow z = 2(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}))$

⑤ Exponential Form
Taylor form: $e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \dots = \cos \theta + i \sin \theta$
Euler's Formula: $e^{i\theta} = \cos \theta + i \sin \theta$
 $\Rightarrow |e^{i\theta}| = |\cos \theta + i \sin \theta| = 1$
 $e^{i\theta} = e^{i(\theta)} = \cos(\theta) + i \sin(\theta) = \cos \theta + i \sin \theta$

Hence: $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$
 $\Rightarrow z = r(\cos \theta + i \sin \theta) = re^{i\theta}$ "Exponential Form"

if $z_1 = r_1 e^{i\theta_1}, z_2 = r_2 e^{i\theta_2}$
① $z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$ ② $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$
Remark: let $w = re^{i\theta}$
① $w \cdot i = re^{i\theta} \cdot e^{i\frac{\pi}{2}} = re^{i(\theta + \frac{\pi}{2})}$ multiply $\rightarrow$ rotate 90° counterclockwise
② $\frac{w}{i} = re^{i\theta} \cdot e^{-i\frac{\pi}{2}} = re^{i(\theta - \frac{\pi}{2})}$
③ $w(-1) = re^{i\theta} \cdot e^{i\pi} = re^{i(\theta + \pi)}$ multiply $\rightarrow$ rotate 180° counterclockwise

Write $1-i$ in exponential form, then find $(1-i)^{1/2}$ geometrically.
 $r = \sqrt{1^2 + 1^2} = \sqrt{2}$, $\theta = \arctan(-1) = -\frac{\pi}{4}$ since $(1,-1)$ is in 4th quadrant
 $1-i = \sqrt{2} e^{-i\frac{\pi}{4}}$
 $\sqrt{1-i} = \sqrt[4]{2} e^{-i\frac{\pi}{8}}$ Multiply by i corresponds to a rotation by 90° counterclockwise
 $\therefore \sqrt{1-i} = \sqrt[4]{2} e^{i\frac{7\pi}{8}}$

⑥ Roots of Complex Number
if $z = re^{i(2\pi k/n)}$, then its n th roots $(\frac{r}{n})^{1/n}, r^{1/n} e^{i(2\pi k/n)}$, $k=0,1,\dots,n-1$
ex. Find both square roots of i :
 $i = 1 \cdot e^{i(\frac{\pi}{2} + 2\pi k)}$, $k=0,1$
 $(i)^{1/2} = e^{i(\frac{\pi}{4} + \pi k)}$, $e^{i(\frac{\pi}{4} + 2\pi)}$, $k=0,1$
when $k=0$: $e^{i(\frac{\pi}{4} + 2\pi)} = e^{i\frac{\pi}{4}}$
when $k=1$: $e^{i(\frac{\pi}{4} + 2\pi)} = e^{i\frac{9\pi}{4}}$
 $\therefore (i)^{1/2} = e^{i\frac{\pi}{4}}$ or $e^{i\frac{9\pi}{4}} = -e^{i\frac{3\pi}{4}}$

The n th root of unity
if $z^n = 1$, since $1 = e^{i \cdot 0}$, $e^{i \cdot 2\pi}$
 $z_k = e^{i \frac{2\pi k}{n}}$, $k=0,1,\dots,n-1$

ex. Find 3 cube roots of unity
 $z_k = e^{i \frac{2\pi k}{3}}$, $k=0,1,2$
 $\Rightarrow z_0 = e^{i \cdot 0} = 1$, $z_1 = e^{i \frac{2\pi}{3}}$, $z_2 = e^{i \frac{4\pi}{3}}$

The roots are always equally spaced in space.

Connection to differential equation
For $a, b, c \in \mathbb{R}$, $e^{(a+bi)t} = e^{at} \cdot e^{bit} = e^{at} (\cos(bt) + i \sin(bt))$
 $= e^{at} \cos(bt) + i e^{at} \sin(bt)$ (with t)
"a" controls growth/decay, $a > 0$: growth / $a < 0$: decay
"b" controls oscillation, $b \neq 0$ speed $\uparrow$

1.2: Introduction to ODEs and First-Order Methods

① ODE: Ordinary Differential Equation
Equation of an unknown function of one variable and its ordinary derivatives
eg. $y'' + y' = x$, $5y' + y'' + 6y = \cos x$
② PDE: Partial Differential Equation
Equation of an unknown function of several variables and its partial derivatives
eg. $u_x = k u$
③ Order: order of the highest derivatives
Linear: if can be written as: $a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_0(x)y = g(x)$ (there is no $\ln(y)^k$)

(a) $y' + y^2 = x$ ODE, 1, nonlinear
(b) $y'' + 5y' + 6y = \cos x$ ODE, 2, linear
(c) $u_t = k u$ PDE, 2, linear
(d) $(y')^2 + y = 0$ ODE, 2, nonlinear
(e) $(y')^2 + (y')^2 + (y')^2 = 0$ ODE, 3, linear
(f) Continuous compounding with deposit: $\frac{dS}{dt} = rS + k$ ODE, 1, linear
(g) Chemical reaction: $\frac{dC}{dt} = k(A_0 - C)(B_0 - C)$ ODE, 1, nonlinear

Solutions and Initial Value Problems (IVP)
① Solutions: $\begin{cases} \text{a function: satisfying the equation} \\ \text{An interval: on which the function satisfies the equation} \end{cases}$
② n th-order ODE has a general solution containing n arbitrary constants
Conditions determines these constants
if: at 1 point x_0 , all conditions are prescribed $y(x_0), y'(x_0), y''(x_0), \dots$
Then the problem is an IVP

ex. verify each proposed solution and state its interval restriction
① $y = e^{2x}$ for $y'' - y' - 2y = 0$
 $y' = 2e^{2x}, y'' = 4e^{2x}$, since $4e^{2x} - 2e^{2x} - 2e^{2x} = 0$, $y = e^{2x}$ is a solution
interval: $x \in (-\infty, \infty)$
② $y = c/x$ for $xy' + y = 0$
 $y' = -\frac{c}{x^2}$, $xy' + y = -\frac{c}{x} + \frac{c}{x} = 0 \Rightarrow y = \frac{c}{x}$ is a solution
interval restriction: $(-\infty, 0) \cup (0, \infty)$

The Simplest ODEs
if RHS depends only on the independent variable:
 $\frac{dy}{dx} = G(x) \Rightarrow \begin{cases} \frac{dy}{dx} = G(x), \frac{dy}{dx} = G(x) \Rightarrow \text{No Simplest ODE!} \\ \frac{dy}{dx} = G(x) + f(y) \Rightarrow \text{No Simplest ODE!} \end{cases}$

ex. A ball is dropped from a 50m building, when does it reach the ground?
let $x(t)$ be height above the ground
 $\frac{d^2x}{dt^2} = -g$, $\frac{dx}{dt} = v$, $v - g t = -g t + C$, since $\frac{dx}{dt} = 0$ when $t=0$, $C=0$
 $x = \int -g t dt = -\frac{1}{2} g t^2 + C$, since $x=50$ when $t=0$
 $C=50$, $x = 50 - \frac{1}{2} g t^2$
when $x=0$, $g t^2 = 100$, $t = \sqrt{\frac{100}{g}}$

Euler's method
 $y' = f(x, y)$, $y(x_0) = y_0$ "General First-order IVP"
gives the slope of solution curve, even when no closed-form solution
① Choose a step size $h = \Delta x$
② $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h \cdot f(x_n, y_n)$

ex. Apply Euler's method to $y' = y$, $y(0) = 1$, using $h = 0.25$ until $x = 1$.
 $x_0 = 0, h = 0.25, y_0 = 1$
 $x_1 = 0 + 0.25 = 0.25, y_1 = 1 + 0.25 \cdot f(0.25, 1) = 1.25$
 $x_2 = 0.25 + 0.25 = 0.5, y_2 = 1.25 + 0.25 \cdot f(0.5, 1.25) = 1.5625$
 $x_3 = 0.5 + 0.25 = 0.75, y_3 = 1.5625 + 0.25 \cdot f(0.75, 1.5625) = 1.953125$
 $x_4 = 0.75 + 0.25 = 1, y_4 = 1.953125 + 0.25 \cdot f(1, 1.953125) = 2.44140625$

Separable equation:
A first-order ODE is separable if it can be written as:
 $g(y) \frac{dy}{dx} = f(x)$
 $\Rightarrow g(y) dy = f(x) dx$ For IVP: $y(x_0) = y_0 \Rightarrow \int_{y_0}^y g(y) dy = \int_{x_0}^x f(x) dx$

ex. $y' + \frac{1}{y} = \frac{3}{x}$, $y(1) = 2$
 $y' + \frac{1}{y} = \frac{3}{x} \Rightarrow \frac{1}{y} \cdot y' = \frac{3}{x} - \frac{1}{y}$
 $\int \frac{1}{y} dy = \int \frac{3}{x} dx$
 $\ln|y| = 3 \ln|x| + C$
 $\ln|y| = \ln|x^3| + C$
 $|y| = e^C x^3$

1.3: Linear Equation and Applications

ex. $\frac{dy}{dx} + \frac{2xy}{1+y^2} = y(x) = -1$
 $(1+y^2) \frac{dy}{dx} = 2xy$
 $\int (1+y^2) dy = \int 2xy dx$
 $y + \frac{1}{3} y^3 = x^2 + C$
 $3y + y^3 = 3x^2 + C$
 $y^3 + 3y - 3x^2 = C$
 $y = \frac{-3 \pm \sqrt{9 - 12(1 - 3x^2)}}{2}$
 $= \frac{-3 \pm \sqrt{1 + 48x^2}}{2}$

② Linear first-order IVP has the form:
 $\frac{dy}{dx} + p(x)y = q(x)$, $y(x_0) = y_0$
Multiply $\mu(x)$ since $(\mu y)' = \mu y' + \mu y$
 $\mu y' + \mu p y = \mu q$ let $\mu' = \mu p$ then LHS becomes $(\mu y)'$
 $\frac{d(\mu y)}{dx} = \mu p(x)$
 $\frac{d(\mu y)}{dx} = \mu p(x)$
 $\int \frac{1}{\mu} d(\mu y) = \int p(x) dx$
 $\ln|\mu y| = \int p(x) dx$
 $\mu y = \exp(\int p(x) dx)$
 $y(x) = \frac{1}{\mu(x)} (\mu(x) y(x) + \int \mu(x) q(x) dx)$
 $y(x) = \frac{1}{\mu(x)} (\mu(x) y(x) + \int \mu(x) q(x) dx)$
we can always set $\mu(x_0) = 1$
 $\Rightarrow y(x) = \frac{1}{\mu(x)} (y_0 + \int_{x_0}^x \mu(t) q(t) dt)$
where $\mu(x) = \exp(\int_{x_0}^x p(t) dt)$

ex. $\frac{dy}{dx} + \frac{1}{x} y = \frac{3}{x}$, $y(1) = 2$
 $p(x) = \frac{1}{x}$, $q(x) = \frac{3}{x} \Rightarrow \frac{dy}{dx} + p(x)y = q(x)$
let $\mu(x) = \exp(\int \frac{1}{x} dx) = e^{\ln x} = x$
 $y(x) = \frac{1}{x} (2 + \int_1^x e^{\ln t} \cdot \frac{3}{t} dt)$
 $= \frac{1}{x} (2 + 3 \int_1^x e^{\ln t} \cdot \frac{1}{t} dt)$
 $= \frac{1}{x} (2 + 3 \int_1^x 1 dt)$
 $= \frac{1}{x} (2 + 3(x-1))$
 $= \frac{1}{x} (3x - 1)$
 $y = 3 - \frac{1}{x}$
Also separable equation
 $\frac{dy}{dx} + \frac{1}{x} y = \frac{3}{x}$
 $\frac{1}{x} \frac{dy}{dx} + \frac{y}{x^2} = \frac{3}{x^2}$
 $\int \frac{1}{x^2} dy = \int \frac{3}{x^2} dx$
 $-\frac{1}{x} y = -\frac{3}{x} + C$
 $y = 3 - \frac{1}{x}$

Modeling with first order ODE
1. rate of accumulation = rate in - rate out

ex. Saving with continuous deposits
 $S(t)$: value of an investment, r : annual interest rate
 k : amount deposited per year
 $S(t+dt) - S(t) = rS(t) dt + k dt$
 $\frac{dS}{dt} S(t) = rS(t) + k$
 $\frac{1}{r+k} \frac{dS}{dt} = 1$ dt
 $\frac{1}{r+k} \ln|S(t) + k| = t$
 $\ln|S(t) + k| = (r+k)t$
 $S(t) = e^{(r+k)t} (S(0) + k)$
 $S(0) = 0$
 $S(t) = e^{(r+k)t} (k)$
 $S(t) = k \frac{e^{(r+k)t} - 1}{r+k}$
 $S(65) = \frac{k}{0.06+0.04} (e^{0.1 \cdot 65} - 1) = 800000$
 $k = \frac{800000 \cdot 0.06 \cdot 0.04}{e^{0.1 \cdot 65} - 1} \approx 47889$

Your turn. A 25-year-old starts with no invested capital, earns a real return of 6% per year, and contributes continuously until age 65. How much must be contributed per year to reach HK\$8,000,000?
 $S(t)$: Saving at age t
 $S(t) = 0$
 $S(t+dt) - S(t) = k dt + r S(t) dt$
 $S' = \frac{k}{t} (e^{rt} - 1) = \frac{k}{t} (e^{0.06t} - 1)$
 $S(65) = \frac{k}{0.06} (e^{0.1 \cdot 65} - 1) = 8000000$
 $k = \frac{8000000 \cdot 0.06}{e^{0.1 \cdot 65} - 1} \approx 47889$

Mortgage repayments
 $B(t)$: outstanding loan balance
 r : rate of interest
 k : annual rate of repayment
 $B(t+dt) - B(t) = rB(t) dt - k dt$
 $\frac{dB}{dt} B(t) = rB(t) - k$
 $B(t) = \frac{k}{r} (e^{rt} - 1)$

Your turn. Estimate the monthly installment for a continuous-payment mortgage with $B_0 = \text{HK\$4,000,000}$, $r = 0.06$ and $T = 25$ years.
 $B(t)$: balance
 $0 = B_0 e^{rt} - \frac{k}{r} (e^{rt} - 1)$
 $k = \frac{r B_0}{1 - e^{-rT}} = \frac{0.06 \cdot 4,000,000}{1 - e^{-0.06 \cdot 25}} \approx 240,000$
monthly installment: $\frac{k}{12}$

Chemical Reaction
 $A + B \xrightarrow{k} C$
given $\frac{dC}{dt} = kAB$, $\frac{dA}{dt} = -\frac{dB}{dt} = -kAB$
 $A(t) = A_0 - B(t)$, $B(t) = B_0 - C(t)$
 $C(t) = ?$
 $\frac{d(A+C)}{dt} = \frac{d(B+C)}{dt} = 0$
 $A+C, B+C$ remains unchanged during reaction
 $A(t) = A_0 - B(t) = B_0$
At any time $t \geq 0$, $A = A_0 - C, B = B_0 - C$
At any time $t \geq 0$, $A = A_0 - C, B = B_0 - C$
 $\frac{dC}{dt} = k(A-C)(B-C)$
 $\Rightarrow \int_0^C \frac{1}{(A_0-C)(B_0-C)} dC = \int_0^t k dt$
 $\frac{1}{B_0 - A_0} \ln \left| \frac{C - B_0}{C - A_0} \right| \Big|_0^C = k t$
 $\frac{1}{B_0 - A_0} \ln \left| \frac{(B_0 - C) A_0}{(A_0 - C) B_0} \right| = k t$
 $\frac{A_0(B_0 - C)}{B_0(A_0 - C)} = e^{k(B_0 - A_0)t}$
 $A_0 B_0 - A_0 C - B_0 k t(B_0 - A_0) = B_0 C e^{k(B_0 - A_0)t}$
 $C(A_0 - B_0 e^{k(B_0 - A_0)t}) = A_0 B_0 (1 - e^{k(B_0 - A_0)t})$
 $C = \frac{A_0 B_0 (1 - e^{k(B_0 - A_0)t})}{A_0 - B_0 e^{k(B_0 - A_0)t}}$

1.4: First-Order Application

ex. $\frac{dy}{dx} + y = e^{-x}$, $y(0) = \frac{3}{4}$
 $\mu(x) \frac{dy}{dx} + \mu(x)y = \mu(x)e^{-x}$
let $\mu(x) = e^x$, $\frac{d(\mu y)}{dx} = \mu(x)e^{-x}$
 $\mu y = \int \mu(x)e^{-x} dx = \int e^x e^{-x} dx = \int 1 dx = x + C$
then $\mu y = e^x \mu$
 $(e^x y)' = e^x$
 $\frac{d}{dx} (e^x y) = e^x$
 $e^x y(x) \Big|_0^t = e^x \Big|_0^t$
 $e^x y(x) = \frac{3}{4} e^x - \frac{1}{4} e^x$
 $y(x) = e^x (1 - \frac{1}{4} e^{-x})$

Terminal velocity
 $v(t)$: v of a falling mass m
 $m \frac{dv}{dt} = -mg - kv$, $v(0) = 0$
 $\frac{dv}{dt} = -g - \frac{k}{m} v$, $v(t) = ?$
 $\frac{dv}{dt} + \frac{k}{m} v = -g$
 $\mu \frac{dv}{dt} + \mu \frac{k}{m} v = -g \mu$
let $\mu = e^{\frac{k}{m} t}$, $\mu v = ?$
 $\mu \frac{dv}{dt} = -g \mu$
 $\int \mu dv = \int -g \mu dt$
 $\mu v = -\frac{mg}{k} (1 - e^{-\frac{k}{m} t})$
 $v = -\frac{mg}{k} (1 - e^{-\frac{k}{m} t})$

Escape Velocity
 $x(t)$: height above ground, r : radius of Earth
 v_0 : initial v , M : mass of Earth
 $F = G \frac{Mm}{(r+x)^2} = -m \frac{dv}{dt}$
 $\frac{GM}{(r+x)^2} = -\frac{dv}{dt}$
Since $g = G \frac{M}{r^2}$ and C is constant,
 $\frac{g}{(1+x/r)^2} = -\frac{dv}{dt}$, $v_0 = \sqrt{2gR}$
Kirchhoff's voltage law says that the applied emf equals the sum of the voltage drops around the loop.
 $V_R + V_C = \mathcal{E} = V_R + RC \frac{dV_C}{dt}$
 $\frac{dV_C}{dt} + \frac{1}{RC} V_C = \frac{\mathcal{E}}{RC}$
 $V_C(t) = 0$
let $\mu(t) = e^{\frac{t}{RC}}$, $\mu V_C = ?$
 $\frac{1}{\mu} \frac{d(\mu V_C)}{dt} = \frac{\mathcal{E}}{RC}$
 $\int \frac{1}{\mu} d(\mu V_C) = \int \frac{\mathcal{E}}{RC} dt$
 $\ln|\mu V_C| = \frac{\mathcal{E}}{RC} t$
 $\mu V_C = e^{\frac{\mathcal{E}}{RC} t}$
 $V_C = \mathcal{E} (1 - e^{-\frac{t}{RC}})$

Electrical laws
The component relations are:
 $V_C = \frac{Q}{C}$, $V_R = IR$, $i = \frac{dQ}{dt}$
Since $q = CV_C$ and C is constant,
 $i = \frac{dQ}{dt} = C \frac{dV_C}{dt} = C \frac{dV_C}{dt}$, $V_R = IR = R \frac{dQ}{dt}$
Kirchhoff's voltage law says that the applied emf equals the sum of the voltage drops around the loop.
 $V_R + V_C = \mathcal{E} = V_R + RC \frac{dV_C}{dt}$
 $\frac{dV_C}{dt} + \frac{1}{RC} V_C = \frac{\mathcal{E}}{RC}$
 $V_C(t) = 0$
let $\mu(t) = e^{\frac{t}{RC}}$, $\mu V_C = ?$
 $\frac{1}{\mu} \frac{d(\mu V_C)}{dt} = \frac{\mathcal{E}}{RC}$
 $\int \frac{1}{\mu} d(\mu V_C) = \int \frac{\mathcal{E}}{RC} dt$
 $\ln|\mu V_C| = \frac{\mathcal{E}}{RC} t$
 $\mu V_C = e^{\frac{\mathcal{E}}{RC} t}$
 $V_C = \mathcal{E} (1 - e^{-\frac{t}{RC}})$

Let $N(t)$ be a population with per capita birth rate b and death rate d . Over a short time interval Δt , births add to the population and deaths remove from it.
 $N(t+\Delta t) - N(t) = bN(t)\Delta t - dN(t)\Delta t$
 $\frac{dN}{dt} N(t) = N(t)(b-d)$
let $b-d = r$, $\frac{dN}{dt} N(t) = rN(t)$
 $N(t) = e^{rt} N_0$